

Berries reduce postprandial insulin responses to wheat and rye breads in healthy women.



Starch in white wheat bread (WB) induces high postprandial glucose and insulin responses. For rye bread (RB), the glucose response is similar, whereas the insulin response is lower. In vitro studies suggest that polyphenol-rich berries may reduce digestion and absorption of starch and thereby suppress postprandial glycemia, but the evidence in humans is limited. We investigated the effects of berries consumed with WB or RB on postprandial glucose and insulin responses. Healthy females (n = 13-20) participated in 3 randomized, controlled, crossover, 2-h meal studies. They consumed WB or RB, both equal to 50 g available starch, with 150 g whole-berry purée or the same amount of bread without berries as reference. In study 1, WB was served with strawberries, bilberries, or lingonberries and in study 2 with raspberries, cloudberries, or chokeberries. In study 3, WB or RB was served with a mixture of berries consisting of equal amounts of strawberries, bilberries, cranberries, and blackcurrants. Strawberries, bilberries, lingonberries, and chokeberries consumed with WB and the berry mixture consumed with WB or RB significantly reduced the postprandial insulin response. Only strawberries (36%) and the berry mixture (with WB, 38%; with RB, 19%) significantly improved the glycemic profile of the breads. These results suggest that when WB is consumed with berries, less insulin is needed for maintenance of normal or slightly improved postprandial glucose metabolism. The lower insulin response to RB compared with WB can also be further reduced by berries.